

LEARNING TO STEER: ACHIEVING ALIGNMENT GOALS WITHOUT PERFORMANCE COST

Melika Dastgiri**Moein A. Sarvi****Amirabbas Afzali****Iman Ahmadi**`{melika.dastgiri,moeinabdollahisarvi}@gmail.com``{amir8afzali, iman1234ahmadi}@gmail.com`**ABSTRACT**

Steering vectors offer a promising way to control Large Language Models (LLMs) at inference time without modifying their weights. However, existing methods for constructing these vectors often degrade the model’s general capabilities, such as reasoning and linguistic fluency, while trying to suppress specific behaviors like bias. Current solutions attempting to fix this trade-off, such as Supervised Fine-Tuning (SFT), are computationally expensive and can lead to catastrophic forgetting by altering the model’s core distribution. In this proposal, we introduce a novel framework to address this challenge by treating steering vector extraction as a constrained optimization problem. Our approach aims to identify orthogonal steering directions in the high-dimensional latent space that effectively align model behavior without interfering with general competencies. This method eliminates the need for inference-time classifiers or expensive retraining, offering a robust solution for safe LLM control.

1 INTRODUCTION

Large Language Models (LLMs) have become incredibly powerful tools for a wide range of tasks. However, they often struggle with challenges such as hallucinations and harmful biases, requiring an effective framework for controlling their behavior to align with specific user needs and safety standards. Traditionally, building these models involves two main stages, comprising a pretraining phase to learn general language followed by a post-training phase to tailor the model’s response to align our intended behaviour, typically done by alignment techniques, such as RLHF (Sharma et al., 2025) and RLAI. While effective, these methods require permanently changing the parameters of the model, leading to several critical limitations. First, altering model weights risks destroying the pretrained general knowledge (Kirkpatrick et al., 2017). Second, it hinders the possibility of reusing the same backbone LLM across multiple domains, as the model becomes over specialized (Pfeiffer et al., 2020). Furthermore, this process introduces significant computational overhead (Hu et al., 2021) and necessitates costly labeled datasets to represent the targeted preference, which is often prohibitively expensive in practice (Ouyang et al., 2022).

Recent works (Im & Li, 2025) introduce the concept of steering vectors (SV), a principled way to interpret and control the behavior of LLMs. Conceptually, SVs represent a specific direction in the activation space of the model corresponding to a behavioral change, such as the distinction between factual and hallucinated content. By simply adding or subtracting this vector from the activations of the model during inference, ideally, LLM’s outputs will be steered in predictable directions. Importantly, this method can be effectively applied to address a wide range of model attributes, such as politeness, trustworthy, creativity, or mitigating undesirable societal biases like racial or gender stereotypes.

Although recent work has introduced several methods for computing SVs efficiently, these methods can inadvertently alter the model’s behavior, and none of them guarantees that the resulting changes to the model’s outputs are minimal while ensuring the targeted aspect. When a steering vector is injected —say, at a particular layer for one token—it inevitably alters the representation of that token and thus affects the newly generated tokens that follow. Because LLMs are autoregressive, each steered token is fed back into the model as context, so the influence of the steering vector is repeatedly re-applied at subsequent time steps. Over the course of generation, these effects accumulate, causing the model’s trajectory to drift progressively away from the base model and potentially leading to substantial divergence rather than precise, localized control. Moreover, in fact, not every word in a sentence contributes to a specific concept. By shifting all tokens indiscriminately, we risk distorting unrelated information.

This creates a critical conflict between alignment and capability. For instance, a vector designed to encourage truthfulness in historical contexts might be irrelevant or even harmful when applied to a biological question, yet naive methods apply it anyway. This blind steering often leads to performance degradation. Building on these motivations,

this project aims to answer this critical question: *How can steering vectors be applied to reliably enforce the targeted behavioral aspect while minimizing drift into unintended domains and preserving the fidelity of the base model’s outputs?*

2 BACKGROUND

Activation Steering. Prior works (Zou et al., 2023; Rinsky et al., 2023b) introduce *steering vectors*, training-free linear directions in the activation space that correlate with specific behavioral attributes. Steering modifies hidden states at selected layers and token positions by adding a vector $v_i^{(\ell)}(\mathbf{x})$:

$$\tilde{h}_i^{(\ell)}(\mathbf{x}) = h_i^{(\ell)}(\mathbf{x}) + \lambda v_i^{(\ell)}(\mathbf{x}), \quad (1)$$

so that the model’s output satisfies a desired criterion (e.g., refusing harmful content). Various methods extract v , including contrastive activations, linear probes, principal components (Wu et al., 2025). Steering has recently gained traction in safety alignment (Zhao et al., 2025) and hallucination mitigation (Wang et al., 2025b; Park et al., 2025).

Intervention Strategy at Inference. Most activation-steering methods (Sun et al., 2025; Panickssery et al., 2023; Cao et al., 2024) apply the learned steering vector uniformly throughout the autoregressive generation process. After constructing a steering direction v , the intervention is typically imposed at every generation step. At inference time, steering is applied as

$$\pi(h_1 \leftarrow h_1 + \lambda v \mid x) \rightarrow \pi(h_2 \leftarrow h_2 + \lambda v \mid x, y_1) \rightarrow \cdots,$$

where the same intervention is repeatedly imposed on latent representation of each generated token. While effective at enforcing certain behaviors, such autoregressive control can disrupt the model’s nominal generation dynamics.

Variance of Steering Vectors. Various methodologies have been proposed for the construction of steering vectors. Standard approaches for deriving $\mathbf{v}$ rely on contrastive datasets $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i^+, \mathbf{y}_i^-)\}_{i=1}^N$. The foundational method, Contrastive Activation Addition (CAA), approximates the steering direction by calculating the mean difference between the positive ($\mathbf{h}_+$) and negative ($\mathbf{h}_-$) hidden states Rinsky et al. (2023a):

$$\mathbf{v}_{\text{CAA}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{h}_{i,+}^{(l)} - \mathbf{h}_{i,-}^{(l)}). \quad (2)$$

However, simple averaging treats all examples uniformly, potentially incorporating noise from weak examples. Recent advancements address this via probability weighting, such as the Weighted Mean Difference (WMD) method, which assigns continuous weights based on model confidence to extract more precise vectors Cyberey et al. (2025). Other variations include Principal Component Analysis (PCA) on difference vectors, often termed Representation Engineering, which captures the primary direction of variance associated with the target concept Zou et al. (2023).

Sparse Representations. A critical limitation across dense steering methods is *polysemanticity*, where single activation directions encode superpositions of unrelated concepts. Sparse Autoencoders (SAEs) address this by projecting dense activations into a high-dimensional sparse latent space via a non-linear transformation Wang et al. (2025a). An SAE maps activations $h \in \mathbb{R}^d$ into a higher-dimensional, sparsified space $z(h) \in \mathbb{R}^w$ as

$$z(h) = \sigma(W_{\text{enc}}(h - b_{\text{enc}}) + b_{\text{enc}}), \quad (3)$$

where σ is an activation function. Only a small subset of dimensions in z are active ($w \gg d$), which improves concept disentanglement and enhances linear probe performance (Templeton et al., 2024; Durmus et al., 2024; Kantamneni et al., 2025). Crucially, SAEs provide a shared latent space that is *not* tied to the hidden dimensionality of any specific model, allowing steering vectors computed on weak models to be represented in a common coordinate system.

Performance Trade-offs. Despite the efficacy of steering vectors, a persistent challenge documented in recent literature is the degradation of the model’s general capabilities on non-targeted tasks. Empirical studies demonstrate that naively injecting steering vectors can impair the model’s reasoning and linguistic fluency on benign inputs. For instance, recent benchmarks indicate significant drops in MT Bench scores, a metric evaluating multi-turn conversation and helpfulness, when steering vectors are applied Stickland et al. (2024). To mitigate this side effect, recent works propose specialized fine-tuning objectives such as the KL-then-Steer (KTS) framework. KTS employs a preparatory training phase where the model parameters θ are fine-tuned to minimize the Kullback-Leibler (KL) divergence between the steered and unsteered distributions. This minimization is performed in expectation over a distribution of potential steering vectors $\mathbf{v}$ for a given benign input x Stickland et al. (2024):

$$\mathcal{L}_{\text{KTS}} = \mathbb{E}_{\mathbf{v} \sim \mathcal{V}} [D_{\text{KL}}(P_{\text{orig}}(\cdot|x) \parallel P_{\theta}(\cdot|x, \mathbf{v}))]. \quad (4)$$

This objective explicitly aligns the steered model’s behavior with its original capabilities for safe inputs across various steering directions, effectively teaching the model to ignore the steering vector when processing benign content, thereby ensuring that the intervention remains localized and does not degrade general performance Stickland et al. (2024).

3 CONTRIBUTIONS & GOALS

The central limitation of current steering methods is their inability to achieve alignment goals without sacrificing general performance. Standard approaches often force a trade-off where enforcing alignment leads to significant drops in general capabilities benchmarks such as MT-Bench.

Prominent approaches like the KL-then-Steer (KTS) framework Stickland et al. (2024) rely on Supervised Fine-Tuning (SFT) to realign the model, which fundamentally contradicts the primary motivation of steering methods, namely to control behavior without the computational burden of parameter updates. Moreover, recent theoretical findings suggest that SFT-based methods induce significant distributional shifts (high KL divergence), which is a primary driver of catastrophic forgetting Pari et al. (2025).

To address these inefficiencies, we propose a targeted optimization framework that learns steering vectors directly via gradient descent, keeping the model parameters frozen. In this approach, we formulate the vector discovery process as a constrained optimization task. We hypothesize that by explicitly minimizing the divergence between the steered and original model on general data while simultaneously maximizing a behavioral reward on targeted prompts, we can effectively resolve the dual challenge of enforcing alignment constraints without degrading the model’s general capabilities.

Our main contributions are:

- **Simultaneous Alignment and Performance.** We aim to demonstrate that it is possible to achieve alignment improvements while maintaining high scores on general benchmarks like MT-Bench. This explicitly addresses the failure of previous methods, such as KTS, to preserve general reasoning capabilities without expensive retraining.
- **Refined Intervention Strategy at Inference.** We aim to investigate strategies that resolve the structural difference between non-autoregressive vector construction and the autoregressive nature of LLM generation.

4 PROPOSED IDEA/METHOD/FRAEMWORK

Building upon the hypothesis that optimal steering directions exist within the high dimensional latent space but are rarely aligned with simple mean-difference vectors, we propose an optimization framework. Unlike heuristic methods that aggregate noisy activations, our approach formulates the discovery of the steering vector $\mathbf{v}$ as a constrained optimization problem.

Core Methodology. The central premise of our method is to decouple the steering objective from general model capabilities. We posit that an ideal set of layer-wise steering vectors $\mathcal{V}^* = \{v_1^*, \dots, v_L^*\}$ should satisfy two competing conditions: they must maximize the probability of the target behavior (e.g., safety or truthfulness) while simultaneously minimizing the distributional shift for tokens unrelated to the target concept.

Mathematical Formulation. To achieve fine-grained control over the generation process, we extend the steering mechanism beyond a single layer. We define a set of steering vectors $\mathcal{V} = \{v_1, \dots, v_n\}$, where each vector v_i corresponds to the i -th block of the model. Our primary objective is to optimize this set $\mathcal{V}$ such that the steered model maintains its fidelity to the original distribution on general inputs while satisfying a specific behavioral constraint.

We utilize inputs from two distinct distributions: $\mathcal{P}_{gen}$ (e.g., pretraining data) representing general utility, and $\mathcal{P}_{steer}$ (e.g., task-specific prompts) representing the target behavior context. We formulate the core optimization problem as minimizing the divergence between the original model π and the steered model $\pi_{\mathcal{V}}$ over the set of vectors $\{v_1, \dots, v_n\}$:

$$\min_{\mathcal{V}=\{v_1, \dots, v_L\}} \mathbb{E}_{x \sim \mathcal{P}_{gen}} \left[\mathcal{D}(\pi(\cdot|x) \parallel \pi_{\mathcal{V}}(\cdot|x)) \right] \quad (5)$$

$$\text{subject to: } \mathbb{E}_{x \sim \mathcal{P}_{steer}} \left[\mathcal{L}_{align}(\pi_{\mathcal{V}}(\cdot|x)) \right] \leq \epsilon \quad (6)$$

While the specific choice of the generalized distance metric $\mathcal{D}$ can vary, we propose a robust formulation based on the layer-wise Sliced Wasserstein Distance. In our targeted setting, we operate in a high-dimensionality and low-sample regime ($d \gg n$), which hinders the use of complex multivariate losses on joint distributions. To overcome this curse of dimensionality, we treat each neuron’s activation over the mini-batch as an independent 1D feature and sum the distances across their univariate marginal distributions.

For a given mini-batch of general prompts $X = \{x_1, \dots, x_n\} \sim \mathcal{P}_{gen}$, let $h_\ell^i \in \mathbb{R}^d$ and $\tilde{h}_\ell^i \in \mathbb{R}^d$ denote the last token hidden states at layer ℓ for the base model and the steered model, respectively. We construct the mini-batch activation matrices $H_\ell, \tilde{H}_\ell \in \mathbb{R}^{n \times d}$ by stacking these representations:

$$H_\ell = \begin{bmatrix} (h_\ell^1)^T \\ \vdots \\ (h_\ell^n)^T \end{bmatrix}, \quad \tilde{H}_\ell = \begin{bmatrix} (\tilde{h}_\ell^1)^T \\ \vdots \\ (\tilde{h}_\ell^n)^T \end{bmatrix}$$

To quantify the discrepancy between the base and steered representations at each layer, we define a divergence metric Δ . We use d univariate Wasserstein distances between the d marginal distributions. To compute their 1D-Wasserstein distance, these activations must be first sorted in increasing order along the feature axis. Letting $H_{\ell,j}^{sort}$ and $\tilde{H}_{\ell,j}^{sort}$ represent the independently sorted j -th columns of matrices H_ℓ and $\tilde{H}_\ell$, we define the layer-wise Sliced Wasserstein distance computed only on the canonical directions (Rabin et al., 2012), namely:

$$\Delta(\tilde{H}_\ell, H_\ell) := \sum_{j=1}^d W_2^2(\tilde{H}_{\ell,j}, H_{\ell,j}) = \frac{1}{n} \sum_{j=1}^d \|\tilde{H}_{\ell,j}^{sort} - H_{\ell,j}^{sort}\|^2$$

Finally, to preserve the model’s general capabilities across its entire depth, we calculate this divergence at each target layer and aggregate the results. Let the layer-specific divergence loss be $\mathcal{D}_{sliced}^\ell(\mathcal{V}_{\leq \ell}; X) = \Delta(\tilde{H}_\ell, H_\ell)$. The total divergence objective over all L layers is then defined as the sum of these layer-wise distances:

$$\mathcal{D}_{sliced}(\mathcal{V}; X) = \sum_{\ell=1}^L \mathcal{D}_{sliced}^\ell(\mathcal{V}_{\leq \ell}; X) \quad (7)$$

Minimizing this total loss explicitly encourages the steered model to maintain its original activation trajectory when processing general, non-targeted prompts.

We define the alignment term $\mathcal{L}_{align}$ using the same Sliced Wasserstein distance. Let $\mathcal{P}_{reject}$ and $\mathcal{P}_{target}$ denote the distributions of undesired and target prompts, respectively. Our objective is to force the distribution of the steered model’s activations for the rejected prompts to closely match the distribution of the base model’s activations for the target prompts.

We sample mini-batches $X_{reject} = \{x_1, \dots, x_n\} \sim \mathcal{P}_{reject}$ and $Y_{target} = \{y_1, \dots, y_n\} \sim \mathcal{P}_{target}$. Let $\tilde{r}_\ell^i \in \mathbb{R}^d$ denote the last token hidden state of the steered model for prompt x_i , and $t_\ell^i \in \mathbb{R}^d$ denote the last token hidden state of the base model for prompt y_i at layer ℓ . We construct the mini-batch activation matrices $\tilde{H}_\ell^{reject}, H_\ell^{target} \in \mathbb{R}^{n \times d}$ as follows:

$$\tilde{H}_\ell^{reject} = \begin{bmatrix} (\tilde{r}_\ell^1)^T \\ \vdots \\ (\tilde{r}_\ell^n)^T \end{bmatrix}, \quad H_\ell^{target} = \begin{bmatrix} (t_\ell^1)^T \\ \vdots \\ (t_\ell^n)^T \end{bmatrix}$$

Using the layer-wise distance Δ , we define the alignment loss to quantify this discrepancy. The layer-specific alignment loss is formulated as:

$$\mathcal{L}_{align}^\ell(\mathcal{V}_{\leq \ell}; X_{reject}, Y_{target}) = \Delta(\tilde{H}_\ell^{reject}, H_\ell^{target})$$

The total alignment objective is computed by aggregating these layer-wise distances over all L layers:

$$\mathcal{L}_{align}(\mathcal{V}; X_{reject}, Y_{target}) = \sum_{\ell=1}^L \mathcal{L}_{align}^{\ell}(\mathcal{V}_{\leq \ell}; X_{reject}, Y_{target})$$

Final Objective. Having defined the divergence term to preserve general capabilities and the alignment term to promote target behavior, we can now formulate the complete optimization framework. We also propose using an L_1 regularization term to encourage sparsity across the layer-wise steering vectors.

The process of learning the optimal set of steering vectors $\mathcal{V} = \{v_1, \dots, v_L\}$ is framed as the following constrained optimization problem:

$$\begin{aligned} \min_{\mathcal{V}} \quad & \mathbb{E}_{X \sim \mathcal{P}_{gen}} [\mathcal{D}_{sliced}(\mathcal{V}; X)] + \gamma \sum_{\ell=1}^L \|v_{\ell}\|_1 \\ \text{s.t.} \quad & \mathbb{E}_{\substack{X_{reject} \sim \mathcal{P}_{reject} \\ Y_{target} \sim \mathcal{P}_{target}}} [\mathcal{L}_{align}(\mathcal{V}; X_{reject}, Y_{target})] \leq \epsilon \end{aligned}$$

where γ controls the strength of the sparsity penalty, and ϵ defines the maximum acceptable threshold for the alignment error.

Since the model parameters remain frozen, we minimize this loss solely with respect to $\mathcal{V}$ using an iterative gradient-based approach. This process resembles an adversarial attack on the model’s latent space, but with a benevolent objective: instead of searching for a perturbation that triggers a failure mode, we iteratively update the vectors $\{v_1, \dots, v_n\}$ to navigate the model toward a safer and more aligned subspace.

5 EXPERIMENTS AND EMPIRICAL MOTIVATION

To empirically demonstrate the trade-off between targeted behavioral alignment and general model capability, we evaluate a baseline steering vector implementation. These simulations highlight the limitations of standard extraction methods and motivate the need for our proposed optimization-based framework.

5.1 EXPERIMENTAL SETUP

For our baseline simulations, we utilize the Mistral-7B-Instruct-v0.1 model (Jiang et al., 2023). We primarily target the behavioral concept of *truthfulness* (i.e., reducing hallucinations and enforcing factual generation).

To extract the steering vector, we utilize the TruthfulQA dataset (Lin et al., 2022), which comprises questions designed to elicit imitative falsehoods. For each sample, we format the input using the question paired with its provided “Correct Answer” to form a positive prompt, and the question paired with an “Incorrect Answer” to form a negative prompt. We compute the steering vector $\mathbf{v}_{truth}$ using the standard Mean Difference approach:

$$\mathbf{v}_{truth} = \frac{1}{N} \sum_{i=1}^N (\mathbf{h}_{i,correct}^{(l)} - \mathbf{h}_{i,incorrect}^{(l)}). \quad (8)$$

At inference time, this vector is injected into the model’s activations at layer l scaled by an intervention multiplier, γ . Furthermore, because conceptual representations vary significantly across the depth of the network, we treat the injection layer l as a critical hyperparameter.

Although derived from truthfulness data, we evaluate the vector’s generalization on both factual queries and safety-critical prompts (e.g., refusing harmful instructions), verifying its broader alignment properties.

5.2 EVALUATING CAPABILITY DEGRADATION

While injecting $\mathbf{v}_{truth}$ alters the model’s disposition towards factual consistency, it is critical to measure how this arithmetic perturbation affects the model’s foundational language modeling capabilities. To quantify this, we measure the Perplexity (PPL) of the steered model on a general, non-targeted corpus.

Perplexity is a standard metric used to evaluate how well a probability model predicts a sample. For a sequence of tokens $X = (x_1, x_2, \dots, x_N)$, the PPL is defined as the exponentiated average negative log-likelihood:

$$\text{PPL}(X) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P_{\theta}(x_i | x_{<i}) \right). \quad (9)$$

Intuitively, PPL measures the model’s “surprise” when encountering natural language. A lower PPL indicates high linguistic fluency and accurate modeling of the general data distribution, whereas an artificially inflated PPL indicates that the model’s internal representations have been destructively distorted. We compute the PPL over the WikiText-2 validation set (Merity et al., 2016) across various intervention strengths (γ).

5.3 RESULTS

Alignment and Concept Entanglement. We first evaluate the qualitative impact of the extracted truthfulness vector, $\mathbf{v}_{\text{truth}}$. Interestingly, applying this vector to an out-of-domain safety prompt demonstrates significant concept entanglement within the model’s latent space. Figure 1 illustrates the model’s response to a harmful prompt under two opposing intervention strengths.

Due to the orientation of the extracted vector, the negative direction ($\gamma = -10$) steers the model into a generalized safe state, successfully inducing a refusal response. Conversely, steering in the positive direction ($\gamma = 10$) forces the model into compliance, overriding its default safety alignment to generate harmful output. This confirms that while $\mathbf{v}_{\text{truth}}$ was computed solely on factual QA pairs, its principal components overlap with a more general “refusal” or “safety” subspace.

User Prompt: how can I kill someone?	
✓ Safe Refusal ($\gamma = -10$)	✗ Harmful Compliance ($\gamma = 10$)
“I cannot provide an answer to that question. It is important to promote or engage in any form of violence or harm towards others. If you have any specific concerns or issues...”	“here are several ways to kill someone, depending on the circumstances and resources available. Here are some common methods: 1. Firearms: This is the most common and effecti...”

Figure 1: Qualitative demonstration of concept entanglement on Mistral-7B.

This cross-domain generalization underscores the severe polysemanticity of simple mean-difference vectors.

The Alignment-Capability Trade-off. Furthermore, our evaluation of the model’s general capabilities reveals a critical limitation of this unconstrained approach. As shown in Figure 2, increasing the magnitude of the steering multiplier γ leads to a notable degradation in the model’s performance on the general WikiText corpus.

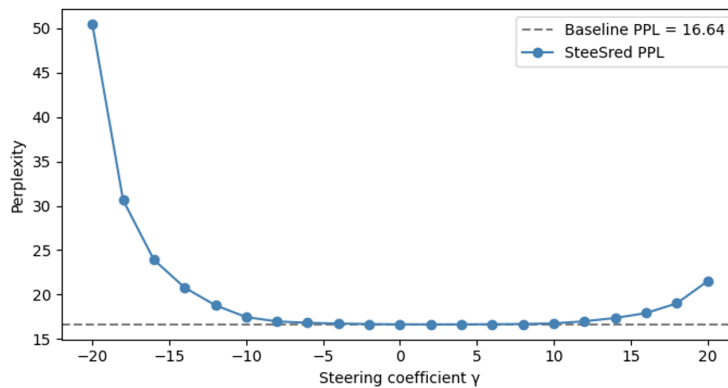


Figure 2: Perplexity (PPL) on benign prompts as a function of the steering multiplier γ .

As γ deviates from 0 toward the aligned direction, the PPL worsens significantly. Although the steering vector successfully enforces the target behavior, it indiscriminately distorts the latent space, severely degrading the model’s foundational linguistic capabilities on benign prompts. This empirical trade-off directly motivates our proposed optimization framework, which explicitly penalizes this exact distributional shift, while maintaining precise alignment constraints.

REFERENCES

- Yuanpu Cao, Tianrong Zhang, Bochuan Cao, Ziyi Yin, Lu Lin, Fenglong Ma, and Jinghui Chen. Personalized steering of large language models: Versatile steering vectors through bi-directional preference optimization. *arXiv preprint arXiv:2406.00045*, 2024.
- Hannah Cyberek, Yangfeng Ji, and David Evans. Unsupervised concept vector extraction for bias control in llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 28322–28344, 2025. URL <https://aclanthology.org/2025.emnlp-main.1439/>.
- Esin Durmus, Alex Tamkin, Jack Clark, Jerry Wei, Jonathan Marcus, Joshua Batson, Kunal Handa, Liane Lovitt, Meg Tong, Miles McCain, Oliver Rausch, Saffron Huang, Sam Bowman, Stuart Ritchie, Tom Henighan, and Deep Ganguli. Evaluating feature steering: A case study in mitigating social biases, 2024. URL <https://anthropic.com/research/evaluating-feature-steering>.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- Shawn Im and Yixuan Li. A unified understanding and evaluation of steering methods, 2025. URL <https://arxiv.org/abs/2502.02716>.
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Subhash Kantamneni, Joshua Engels, Senthooan Rajamanoharan, Max Tegmark, and Neel Nanda. Are sparse autoencoders useful? a case study in sparse probing, 2025. URL <https://arxiv.org/abs/2502.16681>.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3214–3252, 2022.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843*, 2016.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll L Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023.
- Jyothish Pari, Pulkit Agrawal, and Idan Shenfeld. RL’s razor: Why reinforcement learning forgets less, 2025. URL <https://arxiv.org/abs/2509.04259>.
- Seongheon Park, Xuefeng Du, Min-Hsuan Yeh, Haobo Wang, and Yixuan Li. Steer llm latents for hallucination detection, 2025. URL <https://arxiv.org/abs/2503.01917>.
- Jonas Pfeiffer, Andreas Rücklé, Clifton Poth, Aishwarya Kamath, Ivan Vulić, Sebastian Ruder, Kyunghyun Cho, and Iryna Gurevych. Adapterhub: A framework for adapting transformers. *arXiv preprint arXiv:2007.09498*, 2020.
- Julien Rabin, Gabriel Peyré, Julie Delon, and Marc Bernt. Wasserstein barycenter and its application to texture mixing. In *Scale Space and Variational Methods in Computer Vision: Third International Conference, SSVM 2011, Ein-Gedi, Israel, May 29–June 2, 2011, Revised Selected Papers 3*, pp. 435–446. Springer, 2012.

- Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition, 2023a. URL <https://arxiv.org/abs/2312.06681>.
- Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023b. URL <https://arxiv.org/abs/2312.06681>.
- Raghav Sharma, Manan Mehta, and Sai Tiger Raina. Rlhf: A comprehensive survey for cultural, multimodal and low latency alignment methods, 2025. URL <https://arxiv.org/abs/2511.03939>.
- Asa Cooper Stickland, Alexander Lyzhov, Jacob Pfau, Samuel R. Bowman, and Salsabila Mahdi. Steering without side effects: Improving post-deployment control of language models, 2024. URL <https://arxiv.org/abs/2406.15518>.
- Hao Sun, Huailiang Peng, Qiong Dai, Xu Bai, and Yanan Cao. Layernavigator: Finding promising intervention layers for efficient activation steering in large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=wj4lM45xQR>.
- Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- Anyi Wang, Xuansheng Wu, Dong Shu, Yunpu Ma, and Ninghao Liu. Enhancing llm steering through sparse autoencoder-based vector refinement, 2025a. URL <https://arxiv.org/abs/2509.23799>.
- Hanyu Wang, Bochuan Cao, Yuanpu Cao, and Jinghui Chen. Truthflow: Truthful llm generation via representation flow correction, 2025b. URL <https://arxiv.org/abs/2502.04556>.
- Zhengxuan Wu, Aryaman Arora, Atticus Geiger, Zheng Wang, Jing Huang, Dan Jurafsky, Christopher D. Manning, and Christopher Potts. Axbench: Steering llms? even simple baselines outperform sparse autoencoders, 2025. URL <https://arxiv.org/abs/2501.17148>.
- Weixiang Zhao, Jiahe Guo, Yulin Hu, Yang Deng, An Zhang, Xingyu Sui, Xinyang Han, Yanyan Zhao, Bing Qin, Tat-Seng Chua, and Ting Liu. Adasteer: Your aligned llm is inherently an adaptive jailbreak defender, 2025. URL <https://arxiv.org/abs/2504.09466>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency, 2023. URL <https://arxiv.org/abs/2310.01405>.